Andreas Küpfer

Computational Social Scientist

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My research focuses on (multimodal) political communication, combining computational with traditional methods to address substantial questions on party competition and political behaviour.

Current Position

2022-present **Research Associate**
Technical University of Darmstadt
Institute for Political Science (Christian Stecker)

Education

2022-2026 **Ph.D. Political Science**, University of Mannheim
Supervisors: Christian Stecker and Christian Arnold
Committee: Thomas Gschwend, Marc Ratkovic, Christian Stecker
Thesis: *Multimodal Data and Open Science: Innovations in the Study of Political Communication and Behaviour*

2019-2022 **M.Sc. Data Science**, University of Mannheim
Supervisors: Thomas Gschwend and Denis Cohen
Thesis: *When local context trumps party unity: Using Twitter and NLP to explain geographical heterogeneity in legislators' policy communication*

2015-2019 **B.Sc. Business Information Systems**, University of Applied Sciences Karlsruhe

2017 **ERASMUS Student (Studies Abroad), B.Sc. Informatics**, Metropolia University of Applied Sciences in Helsinki, Finland

Publications

2025 **The sound of party competition: how applause reflects unity, disagreement, and the electoral cycle in parliaments**
A. Küpfer, J. Müller, C. Stecker
West European Politics

2025 **The Politics of Seeking and Avoiding Discourse in Parliament**
E. Koch, *A. Küpfer*
European Journal of Political Research

2024 **Nonrandom Tweet Mortality and Data Access Restrictions: Compromising the Replication of Sensitive Twitter Studies**
A. Küpfer
Political Analysis

2024 **The Role of Hyperparameters in Machine Learning Models and How to Tune Them**
C. Arnold, L. Biedebach, *A. Küpfer*, M. Neunhoeffer
Political Science Research and Methods

Working Papers

Look Who's Confronting: Opposition Status, Gender, and the Far Right
A. Küpfer

Structuring Quantitative Image Analysis with Object Prominence
with C. Arnold

Alignment Helps Make the Most of Multimodal Data
with C. Arnold

Measuring Media Slant through Image Analysis
with C. Arnold, O. Rittmann, M. Torres

A Pipeline for Extracting Data from Videos of Complex Political Events
with M. Holman, T. Simko

When Words Divide: MPs as Polarisation Entrepreneurs on the Digital Civic Platform abgeordnetenwatch.de
with C. Sheldon, C. Arnold

Research Visits and Summer Schools

01/2026-10/2026 **Centre for Artificial Intelligence in Government (CAIG) (Christian Arnold)**, University of Birmingham, UK

10/2024-10/2025 **Data and Methods Unit**, Mannheim Centre for European Research (MZES), University of Mannheim, Germany

06/2024 **Video Research Institute, Center for C-SPAN Scholarship & Engagement (CCSE) (Bryce Dietrich)**, Purdue University, US

02/2024-03/2024 **Centre for Artificial Intelligence in Government (CAIG) (Slava Jankin)**, University of Birmingham, UK

Grants, Certificates, and Awards

2025 **Grant: APSA Political Communication Travel Award, 600$**

2025 **Certificate: TU Darmstadt Higher Education Teaching Certificates: Professional Competence Level (200 work units)**

2024 **Grant: German Research Foundation (DFG) Grant "Analysing Legislative Politics with Aligned Text, Audio, and Image Data" with Christian Arnold and Christian Stecker, 285'000€**

2024 **Grant: APSA Political Communication SPARK Travel Award, 750$**

2023 **Award: Athene Award for Excellent Teaching "Promoting data literacy, strengthening critical thinking, taking individual needs into account, integrating modern tools" (TU Darmstadt), 2'000€**

Teaching and Supervision

Master Thesis Supervisions: 1

Student Assistant Supervisions: 5

2024 **Machine Learning for Social Scientists**, PhD graduate school course, MZES (with Ruben Bach)

2024 **Introduction to Analysis and Comparison of Political Systems**, Undergraduate proseminar, Technical University of Darmstadt

2023 **Quantitative Text Analysis in R**, Postgraduate seminar, Technical University of Darmstadt

2023 **Data Analysis: From Data Crawling to Visualization**, Guest lecture (180 minutes), National University of Kyiv-Mohyla Academy

2023 **Getting started with Python: A how-to guide for social scientists**, Workshop with two sessions à 90 minutes, MZES Social Science Data Lab (with Ruben Bach)

2022, 2023 **Introduction to Quantitative Methods in R**, Postgraduate seminar, Technical University of Darmstadt

2022 **Collection, Management, and Analysis of Twitter Data: Using the Twitter API for Academic Research and BERT**, Workshop (90 minutes), MZES Social Science Data Lab

2017 **IT Security and Operating Systems**, Undergraduate tutorial, University of Applied Sciences Karlsruhe

2016 **Programming II**, Undergraduate tutorial, University of Applied Sciences Karlsruhe

2016 **Introduction to Business Informatics**, Undergraduate tutorial, University of Applied Sciences Karlsruhe

Research Dissemination

Invited Talks

2025 **Data Science Brown Bag Series**, Hertie School, Berlin, Germany

2025 **Social Data Science Seminar**, University of Mannheim, Germany

2024 **Mittelbau Colloquium**, MZES, University of Mannheim, Germany

2024 **Text-as-Data (TADA) Speaker Series**, Virtual

Conferences

2025 **APSA (Main Conference and Political Communication Pre-Conference)**[p], Vancouver, Canada

2025 **COMPTEXT**[p,c,d], Vienna, Austria

2025 **PSA**, Birmingham, UK

2024 **APSA**[p], Philadelphia, US

2024 **EPSA**[p], Cologne, Germany

2024 **COMPTEXT**[p], Amsterdam, Netherlands

2024 **IASGP Annual Conference**[c], Darmstadt, Germany

2023 **IASGP Annual Conference**[p], Mannheim, Germany

2023 **ECPR Standing Group on Parliaments**[p,d,c], Vienna, Austria

2023 **EPSA**[p,d], Glasgow, UK

2023 **COMPTEXT**[p,c], Glasgow, UK

2022 **DVPW Group Comparative Research on Parliaments**[p], Bremen, Germany

2022 **EPSA**[p], Prague, Czech Republic

2022 **PolMeth Europe**[p], Hamburg, Germany

2021 **PolMeth Europe**, Virtual

p = Presenter, d = Discussant, c = Panel Chair/Roundtable

Press

2025 **Applaus-O-Meter: Zwei Fraktionen applaudieren vor allem sich selbst**, Frankfurter Allgemeine Sonntagszeitung (FAS) (with Jochen Müller and Christian Stecker)

2025 **When do politicians engage in discourse – and when do they avoid it?**, LSE EUROPP – European Politics and Policy Blog (with Elias Koch)

2023 **Der Frankfurter Wahlkompass VOTO zur OB-Wahl 2023**, FAZ (with Thilo Dieing and Christian Stecker)

2023 **Wahlrecht ab 16, mehr autofreie Zonen und gebührenfreie Kinderbetreuung? Die Positionen der Kandidierenden zur Oberbürgermeisterwahl in Darmstadt in der Online-Wahlhilfe VOTO**, Darmstädter Echo (with Christian Stecker)

2023 **Wahlkompass bietet Orientierung im Kandidatendschungel (FAZ)**, (with Christian Stecker)

2023 **Viele Köpfe und ein breites Angebot. Die Positionen der Kandidierenden zur Frankfurter Oberbürgermeisterwahl**, FAZ (with Christian Stecker)

Web Publications

2023 **BERT and Explainable AI**, Methods Bites, Blog of the Mannheim Centre for European Research (MZES) Social Science Data Lab

2022 **Collection, Management, and Analysis of Twitter Data**, Methods Bites, Blog of the Mannheim Centre for European Research (MZES) Social Science Data Lab

Software

2022 **triviz**, Triangular visualizations for the significance of pairwise differences from multigroup comparisons (Denis Cohen and *Andreas Küpfer*)

Referee

Computational Communication Research • Computational Linguistics for the Political and Social Sciences at KONVENS (Workshop), 2024 & 2025 • Nature Scientific Data • Political Analysis • ZPol

Service

Co-Organizer/Contributor of the Social Science Data Lab (SSDL) at MZES (since 03/2021) • Co-Organizer of the 2024 ECPR Summer School on Parliaments (08/2024) • Member of the decision committee for the German public-private scholarship (Deutschlandstipendium) (since 11/2023) • Data infrastructure representative (since 11/2023)

Selected Work Experience (Academia)

03/2021-09/2024 **Student Research Assistant at Social Science Data Lab**, Mannheim Centre for European Research (MZES) (Denis Cohen)

10/2020-12/2021 **Student Research Assistant at Chair of Political Science, Quantitative Methods in the Social Sciences**, University of Mannheim (Thomas Gschwend)

09/2021-12/2021 **Student Research Assistant at Project Data Science and Big Data at Schools Prodabi**, Paderborn University (Rolf Biehler)

Selected Work Experience (Industry)

07/2018-02/2020 **Working Student Data Science/Finance**, SAP SE in Walldorf, Germany

02/2018-07/2018 **Internship Custom Solution Development**, Capgemini in Stuttgart, Germany

09/2012-07/2015 **Traineeship Software Developer**, KUMAVISION AG in Tiengen, Germany